ORIGINAL ARTICLE

Apple Peel Unfolding of Archimedean and Catalan Solids

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ABSTRACT

We considered a new treatment for making polyhedron nets referred to as "apple peel unfolding": drawing the nets as if we were peeling off apple skins. We define apple peel unfolding strictly and code a program that derives the sequential selection of the polyhedral faces for a target polyhedron in accordance with the definition. Consequently, the program determines whether the polyhedron is peelable (can be peeled completely). We classify Archimedean solids and their duals (Catalan solids) into perfect (always peelable), possible (peelable for restricted cases), or impossible. The results show that three Archimedean and six Catalan solids are perfect, and three Archimedean and three Catalan ones are possible.

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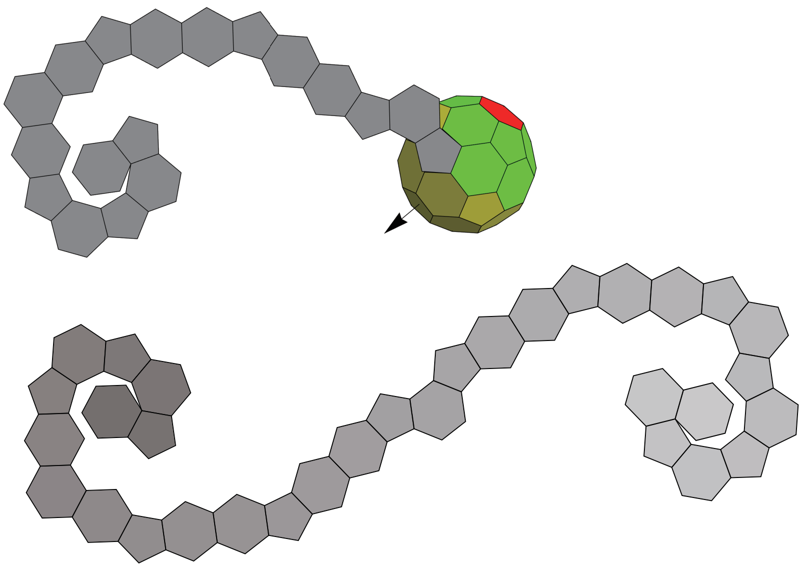

KEYWORDS

Polyhedron net, apple peeling, Archimedean solids, Catalan solids

1. Introduction

Peeling (or unwrapping) an object, such as a fruit or package, is a quite common procedure in our daily lives. Such peeling leads to plane figures in most cases. Unfolding polyhedral surfaces is a well-known subject as a method to obtain polyhedron nets, and has been widely studied from a mathematical point of view, as compiled in Uehara, R (2020). It seems reasonable to consider the peeling of an apple or orange as similar to the unfolding problem of a three-dimensional object. That is, given the axis of an object, we peel off a surface in such a way that the surface is flattened or almost flattened to a plane.

Mathematical consideration of the peeling of an object has been described in various studies as follows. Bartholdi, L. and Heneiques A. (2012) considered peeling off the unit sphere and analyzed the corresponding spiral curve. In the case of polyhedrons, a simplified structure of surfaces, one consideration similar to peeling apples is zipper unfolding, which is slicing a polyhedron along a single cut path like a zipper in a piece of clothing, was proposed by E. D. Demarie et al. (2010), in which a polyhedron net is acquired based on a Hamiltonian path of the polyhedron. Spiral unfolding O'Rourke, J. (2015) expands the surfaces of polyhedrons by cutting their faces spirally.

Apple peeling, the unfolding of a polyhedron such that the polyhedron net is spiral, is an interesting unfolding method of polyhedra. Apple peeling of Platonic solids has already been examined and that consideration was extended to the higher-dimensional polytopes (Kaino, K. , 2019). This procedure can be regarded as sequentially and spirally unfolding the outermost layer of facets with respect to an axis. Furthermore, Kaino demonstrated two results for the dodecahedron, as discussed later. However, there has been no systematic treatment of such unfolding, and a rigorous definition of apple peeling has yet to be clarified.

One application of the polyhedron peeling problem is the study of fullerenes, including the naming of the polyhedrons representing the molecular structure. In particular, the studies of Monolopoulos (Manolopoulos, D.E., et al. , 1991) focused on the peeling of fullerenes, considering them as cubic polyhedrons, together with examples of polyhedra that cannot be peeled (Manolopoulos, D.E. and Fowler, P.W. , 1993). These topics were later investigated and the results significantly updated in the study of Wirz, L.N., et al. (2018) to clarify a general algorithm for finding a face-spiral of a polyhedron.

Our objectives in the present research are (a) to define the apple peeling of a polyhedron strictly and obtain results for Archimedean and Catalan solids and classify those results, and (b) to draw polyhedron nets and planar graphs representing the peelings. The former objective is necessary to extend the topic to arbitrary convex polyhedrons and to higher-dimensional polytopes (e.g., the six regular polytopes in four-dimensional space), although we focus on only three-dimensional polyhedrons in the present study. It should be noted that apple peel unfolding is related to graph theory because peeling polyhedrons is equivalent to finding Hamiltonian paths of the dual polyhedrons. However, solutions to these problems cannot be obtained from their planar graphs directly due to the non-existence of their axis.

The remainder of this article consists of three parts: the definition and the algorithm of apple peel unfolding are described in section 2. The results of numerical simulations of the unfoldings of Archimedean and Catalan solids are explained in section 3. Finally, we discuss the results in section 4.

2. Methods

2.1. *Preliminaries*

For a considered polyhedron $\mathcal{P}$, we denote the sets of faces, vertices, and edges as $F = \{f_1, ..., f_n\}$, $V = \{v_1, ..., v_m\}$, and $E = \{e_1, ..., e_l\}$, respectively. Note that n denotes the number of faces, and f_i consists of k_i integers if the i-th polygon is a k_i-gon; m and l denote the numbers of vertices and edges of the polyhedron, respectively.

It is well-known and fruitful to consider a polyhedron as a graph by associating the set of vertices, edges, and faces of a polyhedron $\mathcal{P}$ with the graph $G = (V, E)$. The degree of a vertex v of V is the number of edges adjacent to v. By Steinitz's theorem, a graph G is a graph of a polyhedron if and only if G is a simple planar graph that is 3-connected. The graph G is called a polyhedral graph or skeleton. Based on the graph information, the Hamiltonian path is a path consisting of a sequence of vertices in which each vertex is visited exactly once.

A net of a polyhedron $\mathcal{P}$ is a planar figure derived by cutting the polyhedral surface. It is called an e-net if the cut is through the edges of a polyhedron. To avoid confusion, we use that the term "net" to mean an e-net in the present article. Figure 1 shows examples of two polyhedrons, a truncated icosahedron (Fig. 1a) and a pentakis dodecahedron (Fig. 1d), with associated polyhedron nets (Figs. 1b and 1e) and polyhedral graphs (Figs. 1c and 1f).

For a given polyhedron $\mathcal{P}$, the dual polyhedron $\mathcal{P}^*$ is the polyhedron for which each face of $\mathcal{P}$ is represented by a vertex of $\mathcal{P}^*$, and each pair of adjacent faces of $\mathcal{P}$ represents an edge connecting vertices of $\mathcal{P}^*$. Given any polyhedron $\mathcal{P}$, its dual can be immediately constructed. The concept of duality is important for two reasons: one is that the polyhedrons we focus on have the duality relation; the other is that the apple-peel unfolding of a given solid is equivalent to the search for the Hamiltonian path of its dual solid with some restrictions.

To specify the solid we are interested in, a polyhedron is said to be an Archimedean solid, or a semi-regular polyhedron, if its faces are regular polygons and the number of faces around each vertex is the same. The dual Archimedean solids are called Catalan solids. We represent the individual Archimedean and Catalan solids by us-

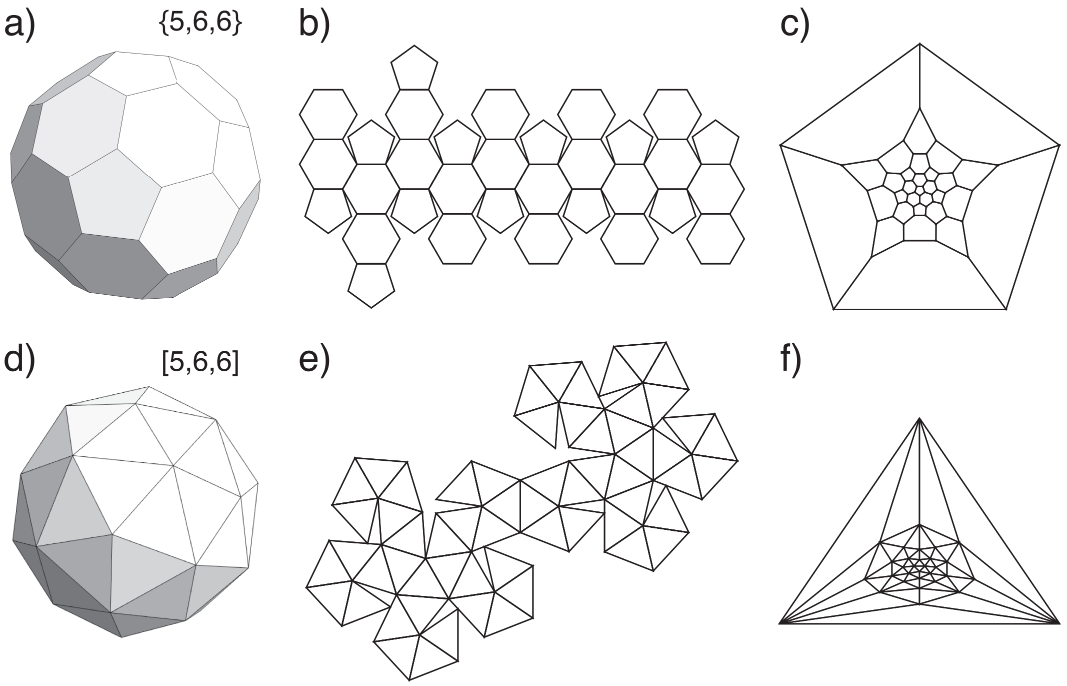

Figure 1. (a) Truncated icosahedron {5,6,6}, (b) net of truncated icosahedron, (c) polyhedral graph of truncated icosahedron, (d) pentakis dodecahedron [5,6,6] (dual of truncated icosahedron), (e) net of pentakis dodecahedron, (f) polyhedral graph of pentakis dodecahedron

ing curly brackets and square brackets, respectively, as follows: $\{n_1, n_2, \cdots, n_q\}$ and $[n_1, n_2, \cdots, n_q]$ refer to Archimedean and Catalan solids, respectively. The former representation indicates each vertex of the polyhedron as being the joint of an n_1-gon, n_2-gon, $\cdots$, and n_q-gon and the latter indicates that each face of the polyhedron has vertices of degrees of n_1, n_2, $\cdots$, and n_q. Therefore, two polyhedrons with a dual relation are represented by the same number combination with different brackets.

Polytope unfolding appears widely in discrete and computational geometry problems. In the present study, we are interested in the unfolding of a polytope following the apple peeling procedure presented in Bartholdi, L. and Heneiques A. (2012) for the case of a sphere, and following Kaino, K. (2019) for the faces of the Platonic solids and the regular polytopes in $\mathbb{R}^4$.

2.2. *Apple Peeling Procedure*

To follow the apple peeling procedure in the real world, we assume that an apple whose skin is peeled off by a knife is rotated along a fixed direction defined by an axis of peeling recognized as the apple stalk. Mathematically, the axis of polyhedron peeling is determined by a fixed ray emanating from the center of the polyhedron while the polyhedron is rotated along a fixed axis direction. Hence, it is necessary to define two other orthogonal rays, which are also orthogonal to the given axis. Remark that the peeling in this context satisfies the unfolding of a polyhedron to a net so that the edges of the polyhedron connect the faces of the polyhedron.

Therefore, we unfold polyhedral faces according to a fixed axis containing the polyhedron's centroid. The selection of all faces is sequentially chosen clockwise (or anti-clockwise) around the axis from top to bottom. The selection's direction corresponds to the handler's dominant hand hereafter. We assume that the handler is right-handed.

Assume that a three-dimensional polyhedron $\mathcal{P}$ is located in the three-dimensional space $\mathbb{R}^3$. A pair of neighboring faces F_1 and F_2 is chosen arbitrarily, and the apple peeling of the facets starts from F_1, and then F_2 is selected as the next face to be peeled off.

For convenience, we set the polyhedron $\mathcal{P}$ so that the origin O is located at the centroid of $\mathcal{P}$. The Cartesian coordinate O-$x_1x_2x_3$ is set so that the x_3-axis contains the centroid of F_1. Then, the axis of peeling of the polyhedron is defined as the x_3-axis, which contains the centroids of both $\mathcal{P}$ and F_1. Specifically, the coordinates of the centroid of F_1 are defined as $(0, 0, x_{3,1})$ and we set the direction of the x_3-axis according to $x_{3,1} > 0$. Therefore, we regard that F_1 is set on the top. As an example of the above definition, we consider the case of the three-dimensional space shown in the schematic illustration of a truncated icosahedron in Fig. 2.

After the faces F_1 and F_2 are selected, the next candidates, the neighboring faces of the previous face in the sequence, are chosen from according to the following priority. If the number of candidates is only one, we use that candidate. If there are multiple candidates, then we select the face whose centroid is at the highest position among the faces on *the left side*. If there are no candidates on the left side, we select the polygon whose centroid is located the lowest among the polygons on the right side. Finally, we continue the selection process until all the faces have been selected or there are no unselected neighbors.

Figure 3 shows an example of the intermediate state of the unfolding procedure of a truncated icosahedron. The arrow shows the x_3-axis of peeling. The planar connection

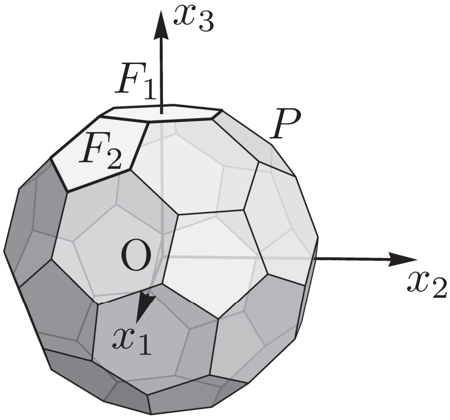

Figure 2. Definition of coordinates in three-dimensional space. The x_3-axis is the axis of peeling.

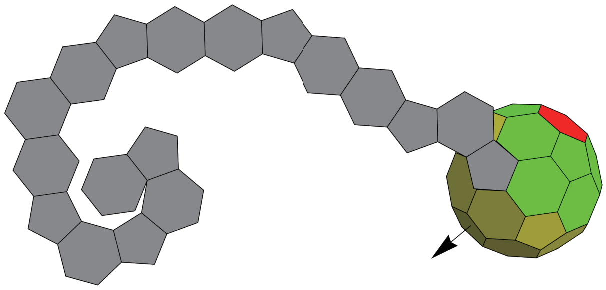

Figure 3. Example of intermediate state of apple peel unfolding of a truncated icosahedron. The arrow represents the axis of peeling.

of gray polygons indicates the intermediate state of its polyhedron net. Yellow and green polygons on the polyhedron are the unfolded and not unfolded faces, respectively. We represent the last selected face with red. The image illustrates that the peeling was started from a hexagon, the handler chose a pentagon as the second polygon, and the peeling continued until the selection of the eighteenth face.

During the peeling procedure, the determination of being on the *'left side'* with respect to the x_3-axis and the considered face F_k needs to be clarified rigorously. Such verification is illustrated in Fig. 4. Suppose that $\mathbf{c}_1$ is a vector proportional to the vector from the origin to c_1, the centroid of face F_1, which is the starting face. For the F_k selected during the procedure with centroid c_k, we construct a vector $\mathbf{c}_k$ as a vector proportional to the vector from the origin to c_k. Then the plane $P_{1,k}$ containing the origin, c_1, and c_k can be defined uniquely and the normal vector of $P_{1,k}$ is obtained as $(\mathbf{c}_k \times \mathbf{c}_1)$, as shown in the figure. To consider whether a point $P(x, y, z)$ is on the left-hand side with respect to the choice of F_k, we verify whether $(\mathbf{c}_k \times \mathbf{c}_1) \cdot \overrightarrow{OP} > 0$,

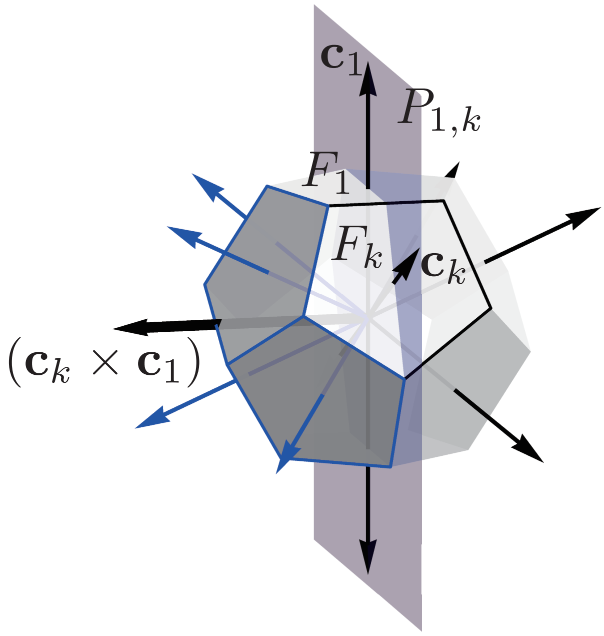

Figure 4. Example of consideration of the left side of a point $P_{1,k}$ with respect to the plane defined by the two faces F_1 and F_k.

i.e., the angle between the vector $(\mathbf{c}_k \times \mathbf{c}_1)$ of plane $P_{i,k}$ and vector $\overrightarrow{OP}$ is acute, as shown by the blue arrows in Fig. 4.

2.3. *Algorithm for Apple Peeling*

The algorithm for numerical simulation is presented as Algorithm 1. Here, a detailed description of the algorithm is given. Input data consist of three sets: (1) vertex coordinates $V = \{v_1, v_2, \cdots, v_m\}$; (2) vertex indices of all faces $F = \{f_1, f_2, \cdots, f_n\}$, where n denotes the number of faces and f_i consists of k integers if the i-th polygon is a k-gon; (3) two neighboring face indexes $\langle F_1, F_2 \rangle$ to start with. The face F_2 must be a neighbor of the first face F_1. The algorithm output is an order of face index, $\langle F_1, F_2, \cdots, F_j \rangle$, and the unfolding is regarded as having finished successfully if $j = n$. We refer to the result in the following as the selection order.

1. Allocate a target polyhedron in the 3D Cartesian coordinate space such that the centroid of the polyhedron is at the origin of the coordinates.
2. Set the counting variable $k = 2$, and rotate the polyhedron to set the centroid of the first polygon F_1 on the top; this means the rotation of the polyhedron sets

the position vector of the centroid of F_1 on the z-axis.

3. Select a face F_{k+1} among the adjacent faces of F_k based on the following priority:
   (a) Select the remaining face automatically as F_{i+1} if it is the only remaining neighbor.
   (b) First, divide the Cartesian space into two parts by the plane composed of the z-axis and the centroid of F_k. Second, list the adjacent polygons whose centroids belong to the left half-space. Finally, select the face which has the largest z-coordinate value as F_{k+1}.
   (c) Select the face as F_{n+1} the face having the z-coordinate which is the highest among the neighboring faces in the left half-space. If there are no neighboring faces in the left half-space, select the neighbor with the lowest z-coordinate.
4. Go back to #3 unless all the faces have been selected or if the next face has no neighboring faces.

For the apple peeling algorithm, we guarantee that the procedure of the algorithm yields a unique sequence by the following lemma.

Lemma 2.1. *Let $\mathcal{P}$ be a polyhedron. The sequence of faces $\langle f_1, f_2, \cdots, f_n \rangle$ obtained by the apple peeling algorithm is unique up to the choice of the initial two faces.*

Proof. We use mathematical induction to prove the argument by starting with $n = 3$. For a given polyhedron $\mathcal{P}$ with n faces; we assume that faces $f_1, ..., f_n$ have centroids $c_1, ..., c_n$, respectively. Let f_α, f_β be the first two faces from the set F, i.e., $f_\alpha = F_1$ and $f_\beta = F_2$, in the sequence generated by Algorithm 1.

We remark that for a given polyhedron $\mathcal{P}$, the centroids of neighboring faces of the face f_k on the left-hand side have different z-coordinates. This is because of the different orientations of faces with respect to the edges of the face f_k affect the positions of the faces' centroids.

For the basis step, we choose the third face F_3 of the sequence satisfying Step 3. Let $f_{\ell,1}, ..., f_{\ell,j}$ be the neighboring faces of face F_2. Then, there exists a plane passing through $\hat{x}_3$ and $\hat{F}_2$, which contains the unique face whose centroid is on the left-hand side of the plane $P_{1,2}$ and is the highest among the neighboring face. Then, the third

Algorithm 1 Algorithm of apple peeling a polyhedron

Require: F_2 must be one of the neighbors of F_1.

function APPLE_PEELING($V = \{v_1, v_2, \cdots, v_m\}$, $F = \{f_1, f_2, \cdots, f_n\}$, $I = \langle F_1, F_2 \rangle$)
  $k \leftarrow 2$
  $l \leftarrow$ a list of unselected neighbors of F_k
  $i \leftarrow$ number of the elements of l
  $c \leftarrow$ list of centroids calculated with V and F
  repeat
    if $i = 1$ **then**
      $F_{k+1} \leftarrow$ the face number listed in l
    else if $i > 1$ **then**
      $q \leftarrow$ number of centroids in the left half
      $s \leftarrow$ list of centroids in the left half
      if $q > 0$ **then**
        $F_{k+1} \leftarrow$ the face number of the face whose z-coordinate of its centroid is the highest among s
      else
        $F_{k+1} \leftarrow$ the face number of the face whose z-coordinate of its centroid is the lowest among s
      end if
    end if
    $k \leftarrow k + 1$
    $l \leftarrow$ a list of unselected neighbors of F_k
    $i \leftarrow$ number of the elements of l
  until $i = 0$
  return $\langle F_1, F_2, \cdots, F_k \rangle$
end function

face can be chosen uniquely.

To prove the inductive step, we assume that the sequence of faces is chosen until the k-th face, i.e., the sequence $\langle F_1, ..., F_k \rangle$ has been generated. The next choice F_{k+1} can be chosen to satisfy the priority in Step 3 for all $k + 1 \leq n$. Since the determination of the left-hand side with respect to the previous face F_k is unique, and the centroids of neighboring faces of F_k are different, the choice of F_{k+1} is also unique. This concludes the proof for the inductive step. Hence, Algorithm 1 gives a unique sequence up to the choice of the two initial faces. □

We remark that the peeling procedure in Step 3 is specified such that the z-coordinates of the face centroids are taken into consideration when faces are chosen. This differs from the study of Wirz, L.N., et al. (2018), which mainly focused on spiral extraction of a polyhedral graph.

2.4. *Peelability*

In the case of peeling an apple or orange in the real world, there exists a continuous curve on a sphere. This curve can be spread out to be a spiral curve on a plane, as stated in the study of Bartholdi, L. and Heneiques A. (2012). Therefore, a similar concept can be considered in the case of a polyhedron. For each polyhedron face F_i, we choose the centroid c_i of the face F_i as a representative point. If two faces F_j, F_k are adjacent, then we assume that there is an edge connecting the two centroids c_j and c_k. Therefore, we can obtain a polygonal (spiral) chain. Since this information can be considered in a discrete structure, it is sufficient to consider it as a sequence of faces from the apple peeling algorithm.

The term *peelable* is defined as the situation where all faces are selected when we apply the apple peeling procedure to a polyhedron. It is determined by the sequential selection of two neighboring faces. By examining all two sequential selections of the neighboring faces, the nature of a polyhedron is determined as either peelable or not. The apple peeling procedure terminates if one of the following conditions is satisfied: no next candidates exist or all faces are in the sequence $\mathcal{F}$, i.e., a polyhedron is successfully peeled or there exists a polyhedral face f_i such that for all shared edges $e_{i,\ell}$ with adjacent faces, $f_\ell \in \mathcal{F}$ but $f_i \notin \mathcal{F}$, which was peeled off incompletely. To clarify rigorously, we would define it based on the sequence generated from the procedure of the apple peeling algorithm with the following.

Definition 2.2. Let $\mathcal{P}$ be a polyhedron with a set of n faces $F = \{f_1, ..., f_n\}$. Suppose that $\mathcal{F}_\alpha = \langle F_1, \cdots, F_t \rangle$ is the sequence of faces generated by the apple peeling algorithm such that $F_i = f_j$ for some $i = 1, ..., t$ and $j = 1, ..., n$, and $F_1 = f_\alpha$ for some α. The polyhedron $\mathcal{P}$ is peelable with respect to the initial face f_α if $|\mathcal{F}_\alpha| = n$. In other words, we say that $\mathcal{F}_\alpha$ is the peelable sequence of $\mathcal{P}$ with respect to the face f_α. Otherwise, it is not peelable with respect to the initial face F_1.

Remark that the existence of a spiral net may depend on the choice of initial faces during the peeling process. We use the term "perfectly" to specify the special properties that it is peelable for any choice of initial faces.

Definition 2.3. (peelability) Let $\mathcal{P}$ be a polyhedron and $\mathcal{F}_\alpha$ be the sequence generated by the apple peeling procedure with the initial face f_α, i.e., $F_1 = f_\alpha$ and $F_1 \in \mathcal{F}_\alpha$.

(1) $\mathcal{P}$ is *perfectly peelable* if $\mathcal{F}_\alpha$ is a peelable sequence for all $\alpha = 1, ..., n$.

(2) $\mathcal{P}$ is *possibly peelable* if there exists a peelable sequence $\mathcal{F}_\alpha$, but there is a sequence $\mathcal{F}_\beta$ which is not peelable.

(3) $\mathcal{P}$ is *non-peelable* if $\mathcal{F}_\alpha$ is not a peelable sequence for all $\alpha = 1, ..., n$.

2.5. *Visualization*

We used three types of visualizations in the following: a) a three-dimensional representation to display the sequential selection, b) a polyhedron net, and c) a planar graph. We have already shown an example of the three-dimensional representation in Fig. 3, along with a description. In addition to the description, we used red polygons to represent the remained ones during the peeling. This means that the existence of red polygons indicates the failure of peeling. Polyhedron nets are used to display the results of the unfolding when it finishes successfully. We used a grayscale to represent the sequence; from dark to light gray corresponds to from a start to a goal of peeling. A planar graph is used to illustrate the whole trace of the selection.

The three-dimensional representation and the polyhedron net use the same procedure: a repetition of rotations of polygons. In the case representing the intermediate state of peeling in three-dimensional space, we rotate the polygons F_k $(k = 1, 2, 3, \cdots, i)$ with respect to the edge shared by F_i and F_{i+1} simultaneously such that the solid angle between the two faces is flat. We obtain a polyhedron net by repeating rotating from $i = 2$ to $i = n - 1$ continuously, where n denotes the number of elements of the order. It should be noted that some disconnections were observed in the case of some polyhedron nets for a Catalan solid due to numerical errors. We corrected such disconnections manually as necessary. The insides of the polyhedrons are facing us for all the polyhedron nets in the following.

The planar graph representation is functional, especially in the case of uniform polyhedrons, because the symmetry properties of the polyhedrons are mostly conserved. First, we project the vertices from $(0, 0, \hat{x}_3)$, $\hat{x}_3 > x_{3,1}$, the xy-plane and arranged the vertices based on the Tutte method (Tutte, W. , 1963). Second, we rotate the vertices around the origin so that the projected centroid of F_2 is on the y-axis. Third, we draw the edges. Finally, we add the numerical results by drawing the lines connecting the centroids of the faces and coloring the unselected (remaining) faces red. Consequently, the outermost polygon of the planar graph represents the exact shape of F_1 and the inner structure of the graph represents the vertex connections.

3. Results

3.1. *Numerical Simulations*

We carried out numerical simulations using Mathematica™. The primary data of polyhedrons, such as the vertex coordinates and vertex numbers of each face, were obtained from a library of Mathematica ("PolyhedronData" function). Although Mathematica allows us to manipulate the exact values (e.g., π and $\sqrt{3}$), we used approximated values to decrease the execution time because the simulation using exact values took an impractical amount of time. This means that the results were numerical rather than exact. We tried all possible combinations of $\langle F_1, F_2 \rangle$ as the initial values of the simulation and summarized the results comparisons of their polyhedron nets.

3.2. *Platonic Solids*

All the Platonic solids were classified as perfect polyhedrons because the Platonic solids are not changed in nature by the selection of F_1 and F_2, so the results are the same for all combinations of F_1 and F_2. Figure 5 illustrates these results, which have previously been reported (e.g., Kaino, K. , 2019). We obtained only one type of polyhedron net for dodecahedrons, although Kaino listed two types of polyhedron nets. This means that only the polyhedron net in Fig. 5 matches our definition of apple peel unfolding. It is notable that we did not obtain the unfolding of a dodecahedron labeled by Kaino as the Atake-type unfolding; the net referred to in Fig. 1 as 'Dodecahedron 1'. The net consists of ten pentagons connected in a zigzag fashion, and the remaining two pentagons opposite each other on the top and bottom.

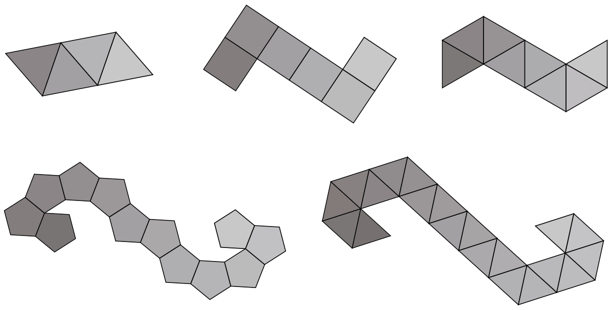

Figure 5. Polyhedron nets of Platonic solids obtained from apple peel unfolding.

3.3. *Archimedean Solids*

The results for Archimedean and Catalan solids are summarized in Fig. 6 and Table 1. Figure 6 illustrates the examples of the polyhedron nets for which the unfolding has finished successfully for all peelable polyhedrons. For Archimedean solids, the numbers of perfect, possible, and impossible polyhedrons are three, three, and seven, respectively. On the other hand, those for Catalan solids are six, three, and four, respectively. There is no apparent relation between the results of Archimedean solids and those of their duals. Therefore, we conclude that the symmetry properties of the faces and vertices do not play a critical role in peelability.

We included information on the existence of the Hamiltonian paths to Table 1 because apple peeling is the problem of finding Hamiltonian paths under restricted conditions. Seven Catalan solids have Hamiltonian paths, as shown in the table. This suggests that some Archimedean solids intrinsically cannot be apple-peeled. We omit the information on the Hamiltonian paths of the Archimedean solid in the table because all solids have Hamiltonian paths. From this viewpoint, the ratios of the number of peelable solids to the number of solids having Hamiltonian paths are 6/7 and 9/13 for Archimedean and Catalan solids, respectively.

All perfect Archimedean solids contain hexagons. The polyhedron nets of a truncated icosahedron {5,6,6} are summarized as the three types shown in Fig. 7 as examples of perfect solids. What differs among the three types is the appearance of the first pentagon; they are $F_1 = 5$, $F_2 = 5$, and $F_3 = 5$ from left to right. The results of the polyhedron were the same as the spiral naming of C_{60} fullerene (Manolopoulos, D.E., et al. , 1991).

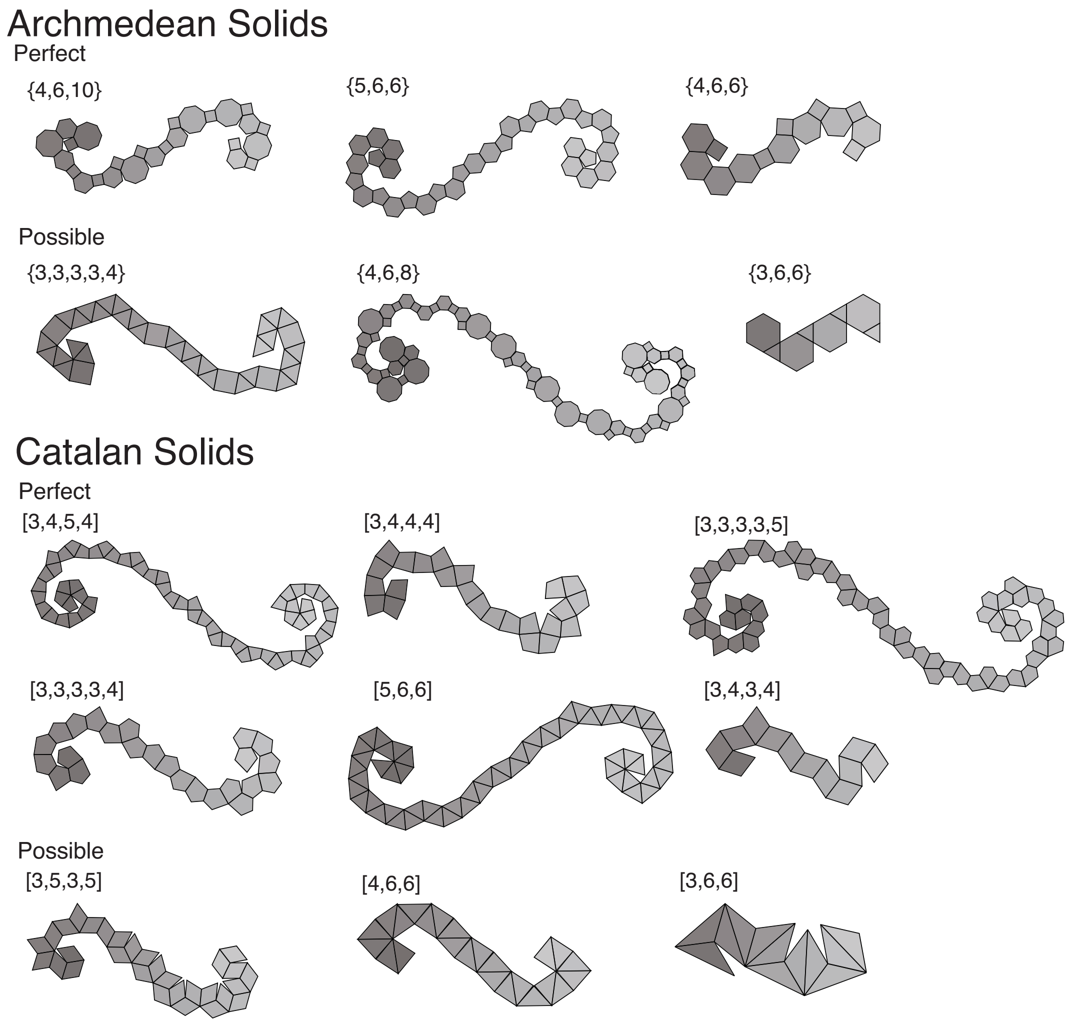

Figure 6. Examples of polyhedron nets of all polyhedrons for which they were successfully completed .

Table 1. Summary of numerical simulations. Bold text indicates that complete results are presented herein.

Name	Index	Peelability Archimedean	Peelability Dual (Catalan)	Hamiltonian path of Catalan solids
Cuboctahedron	{3,4,3,4}	Impossible	Perfect	n.a.
Great Rhombicosidodecahedron	{4,6,10}	Possible	Impossible	available
Great Rhombicuboctahedron	{4,6,8}	Perfect	Impossible	available
Icosidodecahedron	{3,5,3,5}	Impossible	Possible	n.a.
Small Rhombicosidodecahedron	{3,4,5,4}	Impossible	Perfect	n.a.
Deltoidal Icositetrahedron (Rhombicuboctahedron)	{3,4,4,4}	Impossible	Perfect	n.a.
Snub Cube	{3,3,3,3,4}	**Possible**	Perfect	available
Snub Dodecahedron	{3,3,3,3,5}	Impossible	Perfect	available
Truncated Cube	{3,8,8}	Impossible	**Impossible**	n.a.
Truncated Dodecahedron	{3,10,10}	**Impossible**	Impossible	n.a.
Truncated Icosahedron	{5,6,6}	**Perfect**	**Perfect**	available
Truncated Octahedron	{4,6,6}	Perfect	**Possible**	available
Truncated Tetrahedron	{3,6,6}	Possible	Possible	available

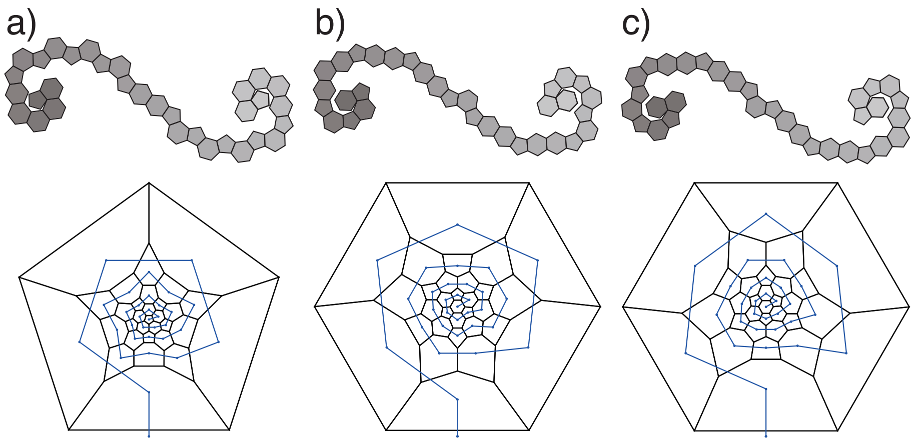

Figure 7. Results of apple peel unfolding of truncated icosahedron {5, 6, 6} (a: $F_1 = 5$, b: $F_2 = 5$, and c: $F_3 = 5$.

The results for a snub cube are summarized in Fig. 8 with polyhedron nets and planar graphs. The three patterns in the top row (Figs. 8a, 8b, and 8c) represent the successful cases, and the rest are the failed cases. The peelings from a square have two patterns (Figs. 8a and 8d), and those from a triangle have four. Chirality affects the results in this case. Comparison of Figs. 8a and 8d shows the influence in the cases starting from a square. The appearance of the second square is one of the differences between the two patterns. The peeling succeeded if the second square appeared as the fourteenth polygon and failed if the square appeared as the sixteenth. In the case of a left-handed person, unfolding succeeded in Fig. 8d and failed in Fig. 8a. There are thirteen triangles between the first and second squares for the peelable cases. On the other hand, the effect appeared in different ways in the cases starting from a regular triangle. The results were classified into four types, illustrated in Figs. 8b, 8c, 8e, and 8f.

The planar graph of Fig. 8b has three-fold rotational symmetry. Therefore, any choice as F_2 among A, B, or C in Fig. 8b gives the same result. Furthermore, it was peelable for left-handed people. On the other hand, the choice among A, B, or C in Fig. 8c affected the result. Only the case of selecting A as F_2 in Fig. 8c was peelable, and Selecting B and C corresponded to Fig. 8e and Fig. 8f, respectively.

It should be noted that a snub dodecahedron, another Archimedean solid having chirality, was an impossible polyhedron. In other words, all the trials of its apple

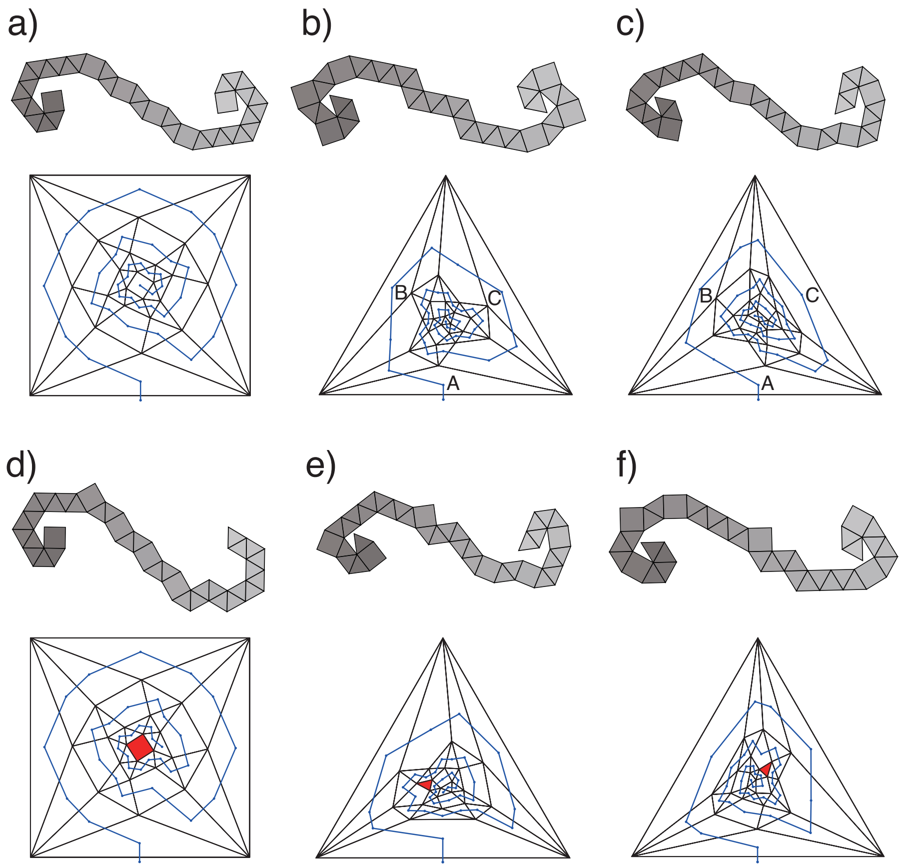

Figure 8. Comparison between a polyhedron net and a planar graph of a snub cube {3,3,3,3,4}.

peeling unfolding failed.

Figure 9 shows the results of truncated dodecahedron {3,10,10}: an example of an impossible polyhedron. We found some variations originated from numerical errors. Although we obtained seven patterns for truncated dodecahedron {3,10,10}, shown in Fig. 9, the basic patterns were only the first three (Figs. 9a, b, and c). The other four results have the same structure as Fig. 9c.

3.4. *Catalan Solids*

Nine polyhedrons were successfully peeled by apple peel unfolding as summarized in Table 1 and Fig. 6. In addition, a comparison between the results and the shapes of the facets is presented in Fig. 10. The polyhedrons having pentagons ([3,3,3,3,4] and [3,3,3,3,5]), kites([3,4,4,4], [3,4,5,4], and rhombuses ([3,4,3,4], [3,5,3,5]) were shown to be perfect. The results of three Catalan solids are presented as examples in Figs. 11,

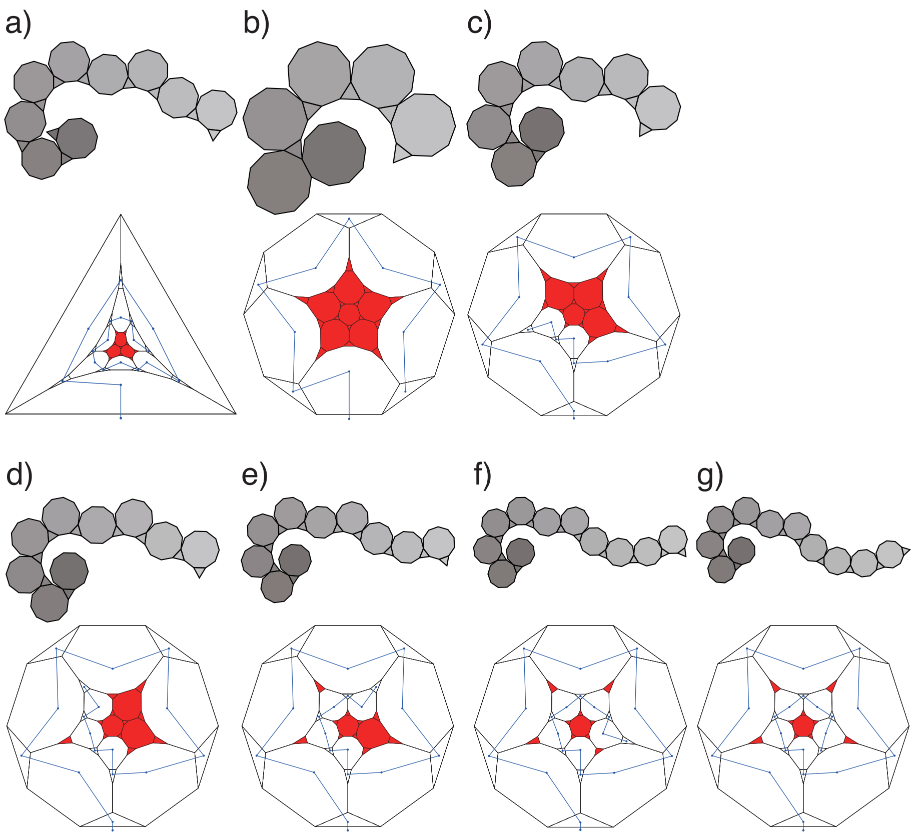

Figure 9. Summary of peelings of truncated dodecahedron {3,10,10}.

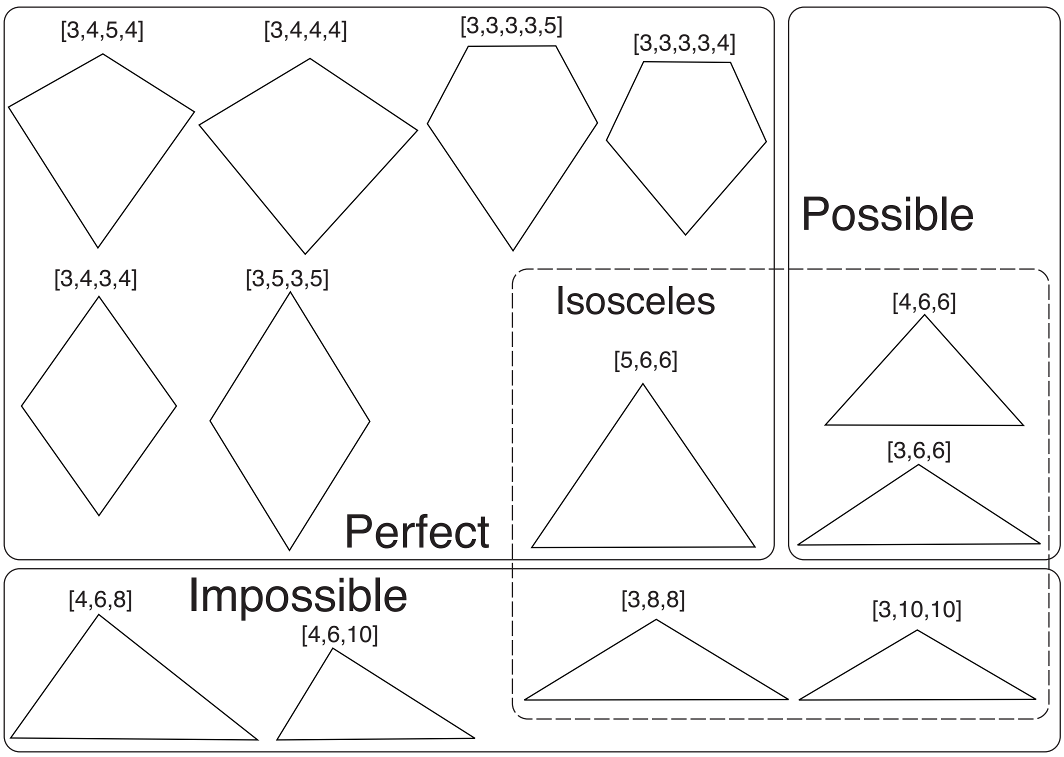

Figure 10. Facets of Catalan solids and their peelability.

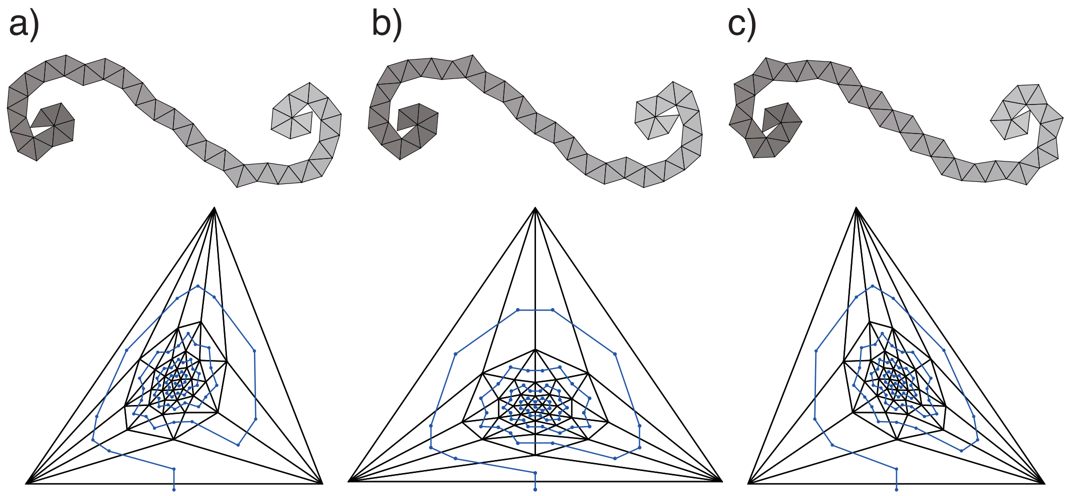

Figure 11. Summary of cases for pentakis dodecahedron [5,6,6].

12, and 13.

All three examples are polyhedrons consisting of isosceles triangles. The pentakis dodecahedron [5,6,6] was classified as perfect: all peelings succeeded. Figure 11 demonstrates three types of unfolding, and the difference arose from the connection of the first two faces. The result of the tetrakis hexahedron [4,6,6], as an example of a possible polyhedron, is shown in Fig. 12. The quadrilateral is the first polyhedron, which differs among the nets. All trials of small triakis octahedron [3,8,8] were classified into three nets , as shown in Fig. 13.

4. Discussion

Using the proposed procedure, we successfully obtained results of apple peel unfolding for Archimedean and Catalan solids. The proposed definition and the algorithm to obtain the results seem appropriate; however, some issues remain to be discussed. These are extensions to general cases, the criterion of peelability, and the possibility of other definitions.

Regarding whether our definition of apple peel unfolding can be extended to general cases, one extension is to apply the definition to random polyhedrons. The present definition seems to be appropriate for random polyhedrons. Another extension is to higher-dimensional polytopes. We expect that the definition is applicable to four-dimensional polytopes because it is consistent with Kaino's apple peeling of four-

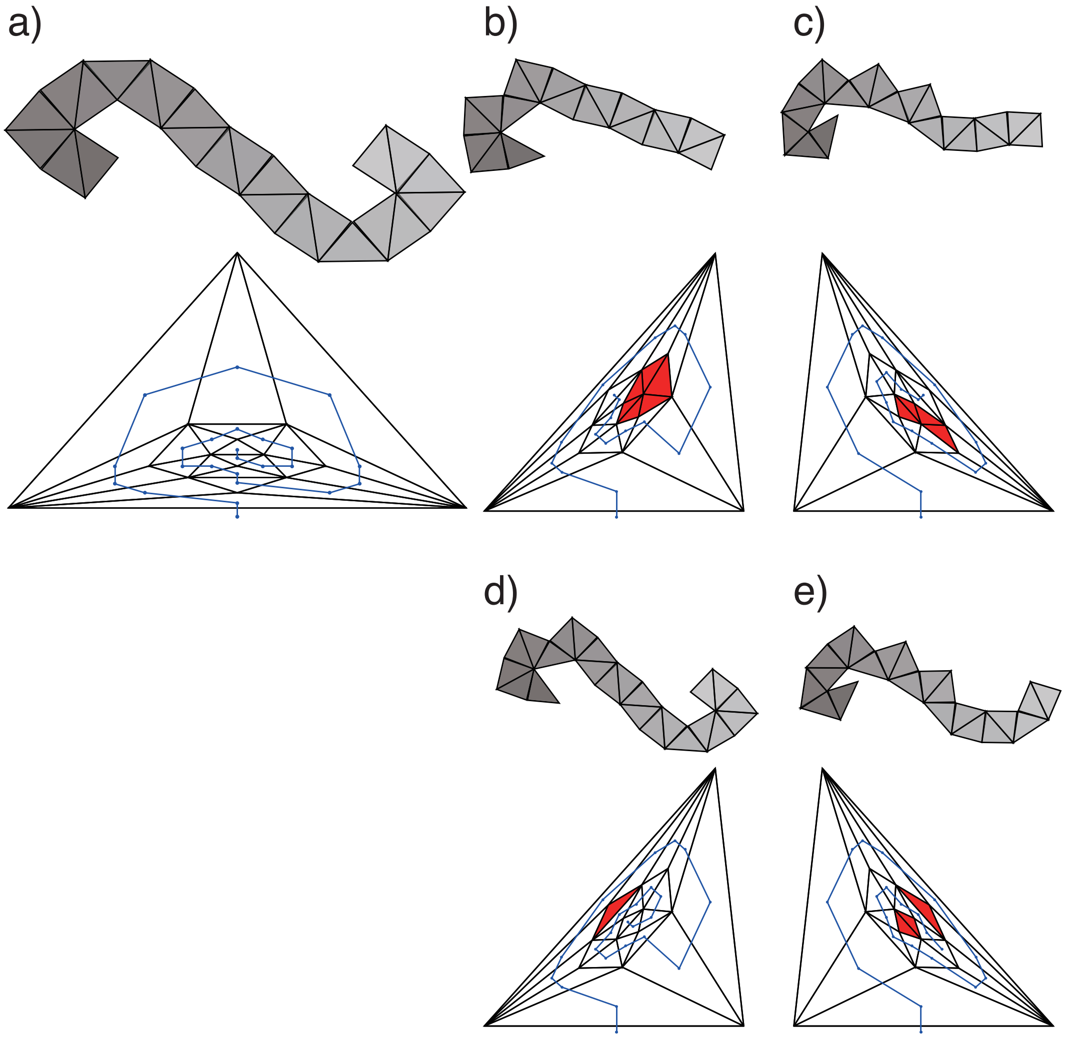

Figure 12. Summary of cases for tetrakis hexahedron [4,6,6]

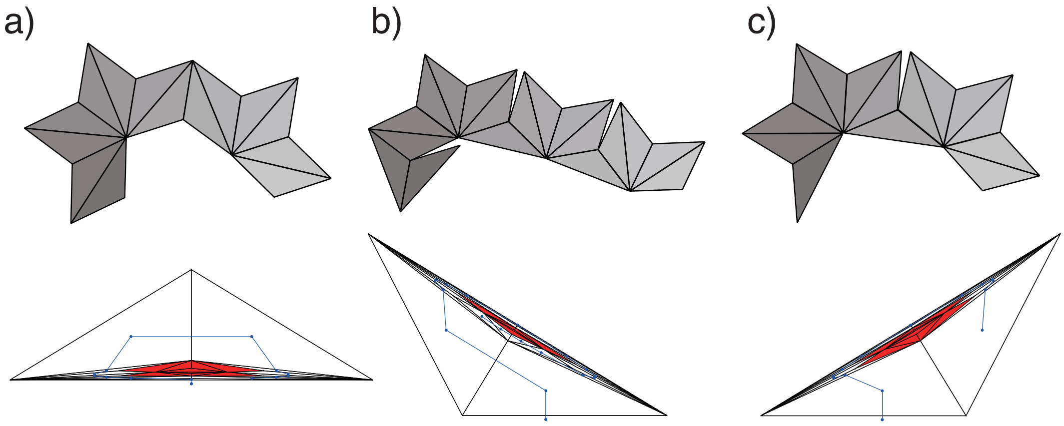

Figure 13. Summary of cases for small triakis octahedron [3,8,8].

dimensional polytopes.

It has not been cleared yet what geometrical property determines the peelability of a given solid. Although we tried to find such a property by considering dualities and Hamiltonian paths, we were not able to. The results for the Catalan solids having triangular faces indicate that a flat shape tends to be not peelable; however, it is not clear how flatness relates to peelability even qualitatively.

Failures of peelings can be classified into two types, although the termination condition is only "no candidate". One is the case that all neighbors have already been selected (termination), and the other is that unselected face(s) were skipped (isolation). Although we could not classify them strictly, the difference will be one of the keys to understanding the results.

Although our definition of apple peel unfolding provided fruitful results, we should discuss the validity precisely because there is another unfolding of a dodecahedron: the Atake-type unfolding. This unfolding could not be obtained in the numerical simulation; however, it can be said that this type is appropriate as another kind of apple peeling. The problem of extending our definition of unfolding to match the Atake-type unfolding remains.

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Disclosure statement

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